

Nathan Ko

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EDUCATION

University of Waterloo

Waterloo, ON

Honours Mathematics/Financial Analysis & Risk Management

Sept 2025 – Present (Expected 2030)

- Intended Minor: Computer Science
- Relevant Coursework: Linear Algebra, Object-Oriented Programming, Functional Programming, Algorithms & Data Abstraction, Probability, Financial Mathematics
- Activities: Varsity Squash Team

EXPERIENCE

Software QA & Test Automation Developer

May 2026 – Present

Government of Ontario

Toronto, ON

- Designed and implemented agentic AI workflows and a reusable prompt library to standardize common QA and testing tasks, cutting repetitive manual effort and improving team-wide productivity and consistency.
- Executed 100+ QA test cases across frontend UI projects within an Agile delivery cycle, validating business requirements and acceptance criteria through functional and regression testing, tracked end-to-end in Azure DevOps.

Software Engineer

Jul 2025 – Aug 2025

CYC Motor Ltd

Hong Kong

- Built and deployed an AI-powered customer support chatbot using a retrieval-augmented generation (RAG) pipeline over internal product manuals, FAQs, and support documentation.
- Designed persona engineering logic to control response tone, reduce hallucinations, and ensure grounded answers from LLMs.
- Integrated the chatbot into the company website, serving 2,000+ monthly user interactions, deflecting a significant portion of human support tickets.

Co-Founder / Lead Software Engineer

Sept 2024 – Present

Aim High Tutoring

West Vancouver, BC

- Founded and scaled a peer-to-peer math tutoring business from 0 to 17 clients in 9 months, generating \$15K+ CAD in revenue with zero external funding.
- Built a client acquisition pipeline with a JavaScript/HTML/CSS landing page integrated with a PostgreSQL (Supabase) database, improving conversion rate by 50%.
- Managed end-to-end business operations across 3 tutors, including hiring, scheduling, client onboarding, and finances.

PROJECTS & EXTRACURRICULAR

Tech Layoff Stock Impact Predictor | *Python, Pandas, Streamlit, scikit-learn, XGBoost*

- Built an end-to-end ML pipeline predicting post-announcement stock drawdown from tech layoff events, using 4,000+ records merged with Yahoo Finance data via Pandas.
- Deployed interactive Streamlit dashboard surfacing model predictions, SHAP feature importance, and historical correlation analysis; evaluated classifier performance via AUC-ROC and precision/recall curves.

Varsity Team Member | *Waterloo Men's Varsity Squash*

- Held the #1 position on the varsity roster, committing 20+ hours weekly alongside a full academic course load.
- Competed at OUA Championships, leading to team's 4th place finish; afterwards placing 13th in U19 national championship.

SKILLS

Languages: Python, SQL, TypeScript, C, HTML/CSS

Frameworks & Tools: Pandas, NumPy, scikit-learn, PostgreSQL, React, Git, Azure DevOps

Spoken Languages: English, Cantonese, Mandarin

Hobbies/Interests: Golf, Running, Weightlifting, Cliff Jumping, Chess, Poker, Philosophy